

supervisor	
signature	

StudentID# : () , Name : ()

* You should write your answer in English. Otherwise, penalty may be applied.

1. (30 points) Fill in the blanks (a)~(t) with the most appropriate English word(s).
- (a. _____) is a programming paradigm based on the concept of (b. _____) which can contain data and code: data in the form of fields (often known as attributes or properties), and code in the form of procedures (often known as methods). In (same as (a)), computer programs are designed by making them out of (same as (b)) that interact with one another.
 - (c. _____) makes extensive use of GOTO statements in order to control the flow of the execution. As a result, the code of (same as (c)) can become very difficult to follow and maintain, especially when it comes to large projects. In (same as (c)), a program contains only a single code block.
 - Using (d. _____) variables is often unsafe because this makes all modules (e. _____) on one another.
 - In C++, (f. _____) is a constructor that is automatically generated by (g. _____) if we do not define any constructor for a class.
 - A pointer pointing to a memory location that has been deallocated (or freed) is called a (h. _____) pointer.
 - In C++, We can define class members (i. _____) using (same as (i)) keyword. When we declare a member of a class as (same as (i)) it means no matter how many (j. _____) of the class are created, there is only one copy of the (same as (i)) member. A (same as (i)) member is shared by all (same as (j)) of the class.
 - In C++, (k. _____) member function does not (l. _____) any data members of the object on which it is called. This restriction is enforced by (m. _____). (same as (k)) member function assures its clients of its benign natures (i.e. safety and readability).
 - (n. _____) and (o. _____) are two key concepts in software engineering that are used to measure the quality of a software system's design. (same as (n)) refers to the degree of interdependence between software modules. (p. _____) (same as (n)) means that modules are closely connected and changes in one module may affect other modules. (q. _____) (same as (n)) means that modules are independent, and changes in one module have little impact on other modules. (same as (o)) refers to the degree to which elements within a module work together to fulfill a single, well-defined purpose. (same as (p)) (same as (o)) means that elements are closely related and focused on a single purpose, while (same as (q)) (same as (o)) means that elements are loosely related and serve multiple purposes.
 - Fred Brooks stated that over 90% of (r. _____) of a typical system arise in (s. _____) phase, and that any successful piece of software will inevitably be (t. _____).
2. (10 points) In C++, it is often desirable to avoid data members in the public area of class. Why? List three advantageous properties we can obtain by avoiding data members in the public area. Fill in the blanks below with appropriate words.
- (i) (a. _____): accessing a data member will be done always through a method (i.e. member function).
 - (ii) precise (b. _____) over the (c. _____) of data members.
 - (iii) functional (d. _____): you can later (e. _____) the data member with a computation.
3. (10 points) (a) What will be the most serious problem if you make every member **private** in a class? Why? Explain with sufficient details.
([Use less than 15 English words] _____).
- (b) What will be the most serious problem if you make every member **public** in a class? Why? Explain with sufficient details.
([Use less than 15 English words] _____).

4.(12 points) Write a program that calculates the size of the file "in.bin" and prints out the calculated file size as output. Insert appropriate C++ code in the following empty boxes. Assume that "in.bin" is a general binary file.

```
#include <iostream>
#include <fstream>
using namespace std;

int main() {
    streampos begin, end;
    [ ] ;
    begin = [ ] ;
    myfile. [ ] ;
    end = [ ] ;
    myfile.close();
    cout << "size: " << (end-begin) << " bytes\n";
    return 0;
}

===== screen output example =====
size: 40 bytes
```

5.(16 points) Write a C++ program that performs file-copying from source binary file "src.bin" to destination binary file "des.bin". Do not use a function (or a command) that performs file-copying operation directly. Instead, you are supposed to read the entire data from the file "src.bin" and write the data to the file "des.bin" for file-copying. Insert your code in the following empty box.

```
#include <iostream>
#include <fstream>
using namespace std;
int main() {
```

```
}
```

6. (22 points) Two lines in the following C++ program contain wrong code where compilation error(s) will occur.

Fill in empty blanks in the following questions. Assume that line 16, line 17, and line 18 are correct.

- (a) What are the two lines that contain the error(s) in the code? : line (), line ()
- (b) Make the code correct by filling in empty blanks below. Note that the logic of the program should be unchanged.
- (i) For correction, line () needs to be modified to ()
- (ii) For correction, line () needs to be modified to ()
- (c) What will be the screen output (i.e. execution result generated by **cout**) of the program after the correction you made in (b)?

Write your answer in the following empty box below.

```
1: #include <iostream>
2: using namespace std;
3: class X {
4: public:
5:     int w,h;
6:     X() { w=0; h=0; cout << "c1\n"; }
7:     X(int a,int b) { w=a; h=b; cout << "c2\n"; }
8:     X(const X& x) { w=x.w; h=x.h; cout << "c3\n"; }
9:     ~X() { cout << "d\n"; }
10: };
11: void func(X obj, int a, int b) {
12:     cout << "f1 called!\n";
13:     obj.w = a; obj.h = b;
14: }
```

```
15: int main() {
16:     X a(5,6);
17:     X* b;
18:     b = &a;
19:     a.w = 2;
20:     func(a,7,8);
21:     X c(a);
22:     b.h = 9;
23:     c.w = 1;
24:     cout << "=====" << endl;
25:     cout << a.w << " " << a.h << endl;
26:     cout << b.w << " " << b.h << endl;
27:     cout << c.w << " " << c.h << endl;
28:     return 0;
29: }
```

screen output
(Write your answer
for (c) here)